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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION, OCTOBER 2025
THIRD SEMESTER
VALUE ADDITION COURSE (VAC) - MG3VACMBG200- MICROBIAL PRODUCTS
(2004 ADMISSION ONWARDS)

Duration : 1 Hour 30 minutes Maximum Marks: 50

PART A
Short Answer Type Questions
Answer any 10 questions
Each question carries 2 marks

1. Mention how plant based milks benefit people with lactose intolerance? [K] / [CO3]

Ans: Any 2 examples for plant based milks (1 mark) Any 2 health benefits (1 marks)

2. Name each of a natural and artificial sweetner? [K] / [CO1]

Ans: Any one example for each. (2 marks)

3. What are skin probiotics [K] / [CO2]

Ans: Definition (1 mark). Any 2 examples (1 mark)

4. Name any two common sources of probiotics [K] / [CO2]

Ans: Yoghurt, Buttermilk, Kefir (Any 2- 2 marks)

5. Define microbial nutrition? [K] / [CO1]

Ans: Definition (1 marks).Name the two main types of nutritional groups of microbes (1 mark)

6. Define hemicelluloses? [K] / [CO1]

Ans: Definition (1 mark) role (1 mark)

7. What makes plant based eggs suitable for vegan? [K] / [CO3]

Ans: Any 2 example for plant based eggs (1 mark). Any 1 health benefit (1 mark)

8. What is skinceutical? [K] / [CO2]

Ans: Definition (1 mark) .Any 2 examples (1 mark)

9. Name two antimicrobial compounds obtained from fermentation [U] / [CO1]

Ans: Nisin, Bacteriocins, Lactic acid (Any 2- 2 marks)

10. Which microorganism converts ethanol to acetic acid in vinegar? [K] / [CO1]

Ans: *Acetobacter aceti* (2 marks)

11. Write one health benefit of drinking kefir. [K] / [CO1]

Ans: Any 2 health benefits 2 marks)

12. Summarize the key features that define cosmoceuticals? [U] / [CO2]

Ans: Definition of cosmoceuticals (1 mark) . Any 2 key features (1 mark)

PART B
Short Essay Questions
Answer any 4 questions
Each question carries 5 marks

13. Write the applicationof skinceuticals in any two skin conditions with example? [K] / [CO2]

Ans: Any 2 application of skinceuticals in any two skin conditions such as acne, hypperpigmentation,anti ageing etc (4 marks) with example (1 mark)

14. Compare mango wine with grape wine? [U] / [CO1]

Ans: Compare raw materials used, sugar content, acid content, fermenting organism, fermentation process, application Any 4 comparisons (5 marks)

15. Explain the most effective ways to educate the public about the benefits and safety of microbial foods? ? [U] / [CO3]

Ans: Health benefits of microbial food (2 marks).Any 3 most effective ways to educate the public.(3 marks).

16. Explain the role of probiotics in improving skin health and preventing skin disorders? [U] / [CO2]

Ans: Definition with example (2 marks). Any 3 role of probiotics in improving skin health and preventing skin disorders (3 marks).

17. Explain the economic and production challenges in the acceptance of microbial food? [U] / [CO3]

Ans: Any 4 challenges (5 marks)

18. Describe any two value added fermented food products? [K] / [CO1]

Ans: Raw materials and microorganism used , nutritional /health benefits of any 2 value added fermented food products such as kefir, Tempeh, yoghurt, Apple cider vinegar(2.5marks each)

PART C
Long Essay Questions
Answer any 1 question
Each question carries 10 marks

19. Define neutraceuticals.Describe their different categories with example? [K] / [CO2]

Ans: Definition (2 marks), Any 6 categories (6 marks), Applications (2 marks)

20. Describe the role of friendly bacteria in the preparation and health benefits of kanji? [K] / [CO1]

Ans: Definition (2 marks), Preparation process (3 marks), Role of friendly bacteria (3 marks), Health benefits (2 marks).